Project Executable Files

Date	02 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This document contains the executable and deployment files of the OptiCrop: Smart Agricultural Production Optimization Engine project. All required source code, configuration files, dependencies, documentation, and deployment details are included to enable the evaluator to run and verify the project successfully.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	N/A
5	Environment configuration file (.env.example or similar)	N/A
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	N/A
8	Dockerfile / Containerization config (if applicable)	N/A
9	Test files and test results	Yes

10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

The following is the file and folder structure of the OptiCrop: Smart Agricultural Production Optimization Engine project. It includes the source code, machine learning model, datasets, templates, static files, and other resources required to build and run the application.

```
OptiCrop/
│
├── dataset/
├── model/
│   └── crop_model.pkl
├── notebooks/
├── screenshots/
├── static/
│   ├── css/
│   ├── images/
│   └── js/
├── templates/
│   ├── home.html
│   ├── about.html
│   └── predict.html
├── app.py
├── crop_data.py
├── requirements.txt
├── README.md
└── .gitignore
```

Step 3: Deployment / Access Details

Field	**Details**
Hosted / Deployed URL	https://opticrop-1-sh40.onrender.com
Login Credentials (Demo)	N/A (No login required)

Platform / Hosting Provider	Render
Repository Link	https://github.com/Mohammed-Sadia-Rahmatunnisa/OptiCrop
Demo Video Link	https://drive.google.com/file/d/1AhqcXDJOe9PhJoyphzc1aduYyOGNXdoO/view?usp=drive_link

Step 4: Run Instructions

Follow the steps below to set up and run the OptiCrop application on a local computer. Ensure that Python and the required dependencies are installed before starting the application.

Pre-requisites:

- Python 3.10 or above
- Visual Studio Code
- Git
- pip (Python Package Manager)

Steps:

```
Step 1: Clone the repository
        git clone https://github.com/Mohammed-Sadia-Rahmatunnisa/OptiCrop.git

Step 2: Open the project folder

Step 3: Install the required dependencies
        pip install -r requirements.txt

Step 4: Run the application
        python app.py

Step 5: Open your browser and visit
        http://127.0.0.1:5000

               or

        https://opticrop-1-sh40.onrender.com
```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Free Render service may take 30–60 seconds to wake up after inactivity.	Working as expected on free hosting plan.
2	Internet connection is required to access the deployed application.	Use the local version (`python app.py`) if offline.
3	Prediction accuracy depends on valid input values.	Users should enter realistic soil and climate values.